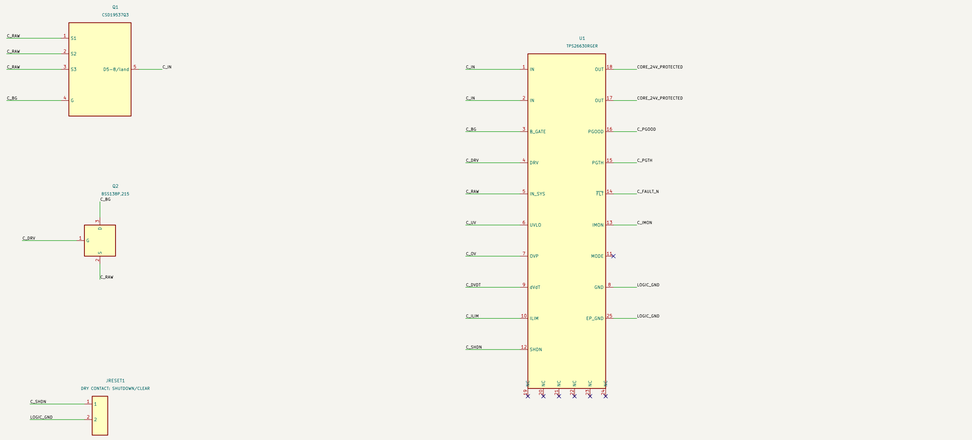
DUAL DC POWER ENTRY — PARTIAL ENGINEERING CIRCUIT r0.2

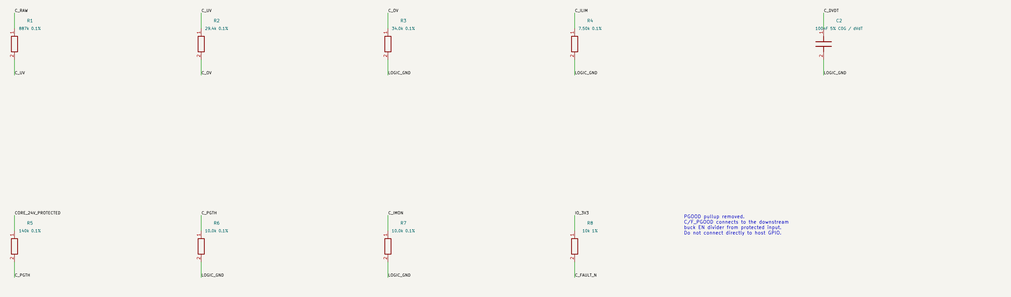
TPS26630 + reverse-blocking FETs; separate LOGIC/FIELD grounds. No PCB or manufacturing release. Exact ratings require the companion review.

CORE / LOGIC DOMAIN

Draft19–30V entry; normal load 1.81A; input path target=≈3A.
Display/AV/motor power excluded. Fuse and wiring selection remains open.







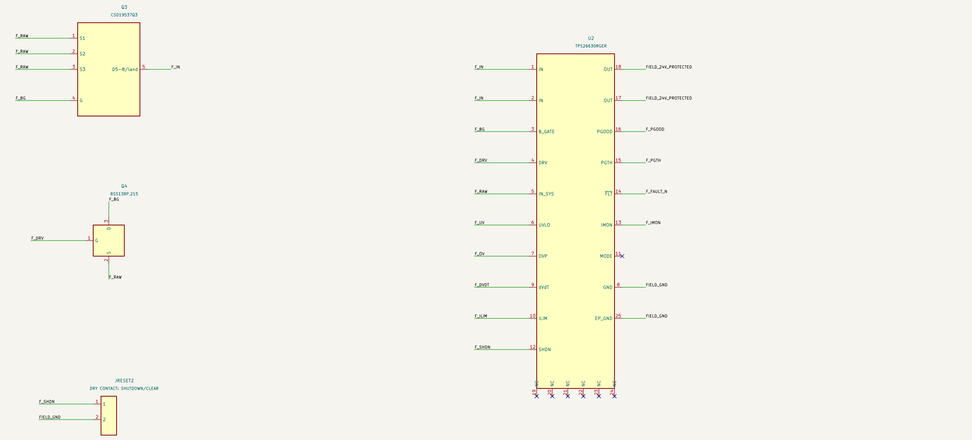
Q1 source faces input; Q2 source also references RAW, not GND.
MODE intentionally open: overcurrent latch-off. Voltage faults can recover.
SHDN floats high internally; dry contact low shuts down/resets the latch.
Power restoration is NOT machine rearm. Permission circuit is separate.

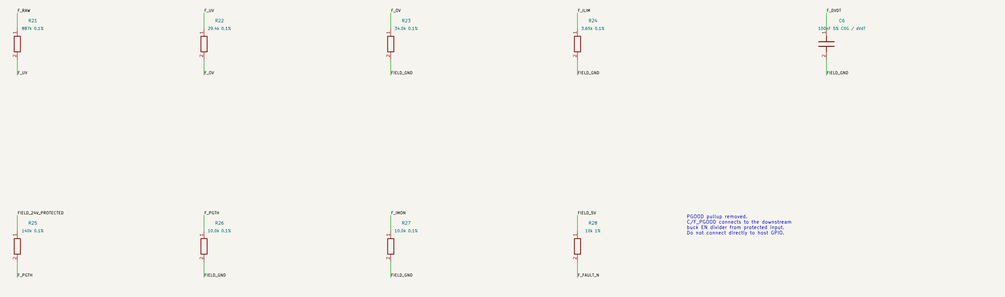
Fault pullups only use downstream diagnostic supply; PGOOD uses converter EN bias.
Effective RAW bypass >=1uF; IN/OUT >=0.1uF. Candidate MPN/bias curves unresolved.
Total downstream capacitance <=1000uF; field loads require independent inhibit.
TVS/clamp/fuse/thermal qualification open. Power flags declare normal paths only.

FIELD DOMAIN — KEEP ISOLATED

Draft19–30V entry; normal load 4.36A; input path target=≈6A.
Display/AV/motor power excluded. Fuse and wiring selection remains open.







Fault pullups only use downstream diagnostic supply; PGOOD uses converter EN bias.
Effective RAW bypass >=1uF; IN/OUT >=0.1uF. Candidate MPN/bias curves unresolved.
Total downstream capacitance <=1000uF; field loads require independent inhibit.
TVS/clamp/fuse/thermal qualification open. Power flags declare normal paths only.

Partial circuit; physical fuse/clamp selection and qualification open

Sheet: /
File: power_entry.kicad_sch

Title: Core and field DC power entry

Size: A1	Date: 2026-09-28	**Rev: r0.2**
KiCad E.D.A. 10.0.4		Id: 1/1